Analytical Mechanics Notes

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1 Introducion

This is a self-made note that summarises the main points from PH2104 Analytical Mechanics, Nanyang Technological University (NTU Singapore). Some of the examples given in the lecture notes were not included as demonstration.

2 Non-Constant Acceleration:Parachute vs Rocket

Mathematically speaking, the two models are all in the form of 1st order ODE:

$$\dot{y}(t) + p(t)y(t) = g(t)$$

Solution to this is have $\mu(t)$ such that $\mu(t)p(t) = \dot{\mu}(t)$, which is $\frac{\dot{\mu}(t)}{\mu(t)} = p(t)$, and it leads to the general solution

$$y(t) = \frac{\int \mu(t)g(t) + c}{\mu(t)}$$

2.1 Parachute Problem

The parachute with mass m is ***damped***. This is to say

$$F^{fr} = k\frac{dx}{dt}$$

This yields Newton's 2nd Law:

$$m\frac{dv}{dt} = mg - kv$$

Solution to this is given, as one can easily prove $\mu(t) = Be^{\frac{kt}{m}}$, $v(t) = \frac{mg}{k} + ce^{\frac{-kt}{m}}$.

2.2 Rocket problem

Key to rocket problem is that we assumed a negative mass δm is cast at a velocity $\boldsymbol{u}$ regarding the rocket while maintaining the conservation of momentum:

$$m\boldsymbol{v} = (m + \delta m)(\boldsymbol{v} + \delta v) - \delta m(\boldsymbol{v} - \boldsymbol{u})$$

Equivalently,

$$m\delta\boldsymbol{v} + \delta m\delta\boldsymbol{v} + \boldsymbol{u}\delta m = 0$$

We neglect the 2nd order term and therefore will yield $\dfrac{dm}{m} = \dfrac{-d\boldsymbol{v}}{\boldsymbol{u}}$.
Naturally, this leads to

$$v_f - v_i = \boldsymbol{u}ln(\frac{m_i}{m_f})$$

To yield maximum momentum, let $\dot{p} = 0$. That is $\dot{m}v + m\dot{v} = 0$, and the maximum momentum is reached at $\dfrac{m_{max}}{m_i} = \dfrac{1}{e}$.

3 Small Oscillations

General principle is to use matrix in describing multi-body systems.
Let equilibrium exist at $x = 0$, therefore expanding a potential function

$$U(x) = \frac{1}{n!}\sum_n x^n \frac{d^n U}{dx^n}$$

Neglecting higher order terms, $U(x) = \frac{kx^2}{2}$. Corresponding force is the restoring force $F = -kx, k > 0$ for stability.
The Newton's 2nd Law to describe this is

$$\boldsymbol{M\ddot{x} = -Kx}$$

in which $\boldsymbol{M}$ is a diagonalised mass matrix, $\boldsymbol{x}$ is the dislocation vector, and $\boldsymbol{K}$ is known as a Hessian matrix, used to describe all the interaction terms:

$$K_{ij} = \frac{\partial^2 U}{\partial_{x_i}\partial_{x_j}}$$

3.1 Simple harmonic motion

Thus for a mass m in SHO, it can be described by $m\ddot{x} = -kx$ whose general solution is $x(t) = Acos(\omega_0 t) + Bsin(\omega_0 t), {\omega_0}^2 = \frac{k}{m}$, exponential expressions are also acceptable.

3.2 Damped Oscillation

"Damped" is usually considered as $f = -k_d\dot{x}$, thus $m\ddot{x} = -kx - k_d\dot{x}$. This is a 2nd Order ODE whose characteristic equation yields solution $\lambda = \frac{-k_d + \sqrt{(k_d^2 - 4mk)}}{2m}$.
Depending on the sign of $\Delta = {k_d}^2 - 4mk$, the system is divided into 3 states:

3.2.1 Overdamped

$\Delta > 0$. This leads to $x(t) = A_1 e^{\lambda_1 t} + A_2 e^{\lambda_2 t}$, as $\lambda_{1,2} < 0$, the displacement x decays rapidly.

3.2.2 Critically damped

$\Delta = 0$. The solution is $x(t) = A_1(1 + kt)e^{\lambda_c t}, \lambda_c = \frac{-k_d}{2m}, k = \frac{A_2}{A_1}$.

3.2.3 Underdamped

$\Delta < 0$. Solution is $x(t) = e^{\frac{-k_d t}{2m}}[c_1 cos(\omega_0 t) + c_2 sin(\omega_0 t)], \omega_0 = \frac{\sqrt{|\Delta|}}{2m}$.

3.3 Input forces and periodic driving

The N2L is: $m\ddot{x} = -kx - k_d\dot{x} + F_0 cos(\omega t)$, a 2nd ordered non-homogeneous ODE. We solve this by $x_g(t) = x_H(t) + X(t)$ as $X(t)$ is the trial solution. In this case the corresponding $X(t)$ is $X(t) = Bcos(\omega t - \phi)$.
After substitution this will yield

$$X(t) = Ae^{\frac{-k_d t}{2m}} cos(\omega_0 t - \alpha) + Bcos(\omega t - \phi), tan\phi = \frac{k_d \omega}{k - m\omega^2}, B = \frac{F_0}{\sqrt{(k - m\omega^2)^2 + (k_d \omega)^2}}$$

3.4 Coupled Oscillators

Here, $\boldsymbol{x}$ is the displacement vector:

$$\boldsymbol{x}(t) = \begin{pmatrix} x_1(t) \\ x_2(t) \\ ... \\ x_n(t) \end{pmatrix}$$

And correspondingly $V(\boldsymbol{x})$ describes the potential energy for the coupled system, and since interactions exist the off-diagonal terms will no longer remain zero. Restoring force is $F_i = -\frac{\partial V}{\partial x_i} = -\Sigma_j K_{ij} x_j$. Normally they each oscillate in different frequencies, but a special solution is when all oscillators have the same frequency known as *normal modes*.

3.4.1 Normal modes

$$\boldsymbol{x}(t) = \begin{pmatrix} A_1 \\ A_2 \\ ... \\ A_n \end{pmatrix} e^{i\omega t}$$

ω is the common frequency. Notice $e^{i\omega t}$ is an eigenvalue to operator $\frac{d}{dt}$ that indicates energy conservation according to Noether's Theorem. This yields

$\ddot{\boldsymbol{x}} = -\omega^2 \boldsymbol{x}$. And the N2L here is reduced to an eigenvalue problem in which $\boldsymbol{M}^{-1}\boldsymbol{K}\boldsymbol{x} = \omega^2 \boldsymbol{x}$.

3.4.2 Example: coupled mass

Consider the example

$$V = \frac{1}{2}k(\Delta x_1)^2 + \frac{1}{2}(2k)(\Delta x_2 - \Delta x_1)^2 + \frac{1}{2}(2k)(\Delta x_2)^2$$

Later Δ is omitted for the sake of simplicity.

$$F_1 = \frac{\partial V}{\partial x_1} = 2kx_2 - 3kx_1; F_2 = \frac{\partial V}{\partial x_2} = 2kx_1 - 4kx_2.$$

To solve this in martix form, $\boldsymbol{M}\ddot{\boldsymbol{x}} = -\boldsymbol{K}\boldsymbol{x}$:

$$\begin{pmatrix} m_1 & 0 \\ 0 & m_2 \end{pmatrix} \begin{pmatrix} \ddot{x_1} \\ \ddot{x_2} \end{pmatrix} = - \begin{pmatrix} 3k & -2k \\ -2k & 4k \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

Substitute $\boldsymbol{x} = \boldsymbol{A}e^{i\omega t}$:

$$\begin{pmatrix} 3-\lambda & -2 \\ -1 & 2-\lambda \end{pmatrix} \begin{pmatrix} A_1 \\ A_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \lambda = \frac{m\omega^2}{k}$$

We'll yield the solution: $\omega_1 = \sqrt{\frac{k}{m}}, \omega_2 = 2\sqrt{\frac{k}{m}}$.

3.4.3 Beats

Consider

$$\omega_1 = \sqrt{\frac{k}{m}}, \omega_2 = \sqrt{(1+2\epsilon)\frac{k}{m}}$$

as the interaction force has $k' = \epsilon k, \epsilon \ll 1$.
Corresponding normal modes are

$$\boldsymbol{A}_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \boldsymbol{A}_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

This leads to

$$\boldsymbol{x}(0) = \begin{pmatrix} d \\ 0 \end{pmatrix} = \frac{c_1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \frac{c_2}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

Such,

$$x_1(t) = \frac{d}{2}(\cos(\omega_1 t) + \cos(\omega_2 t)), x_1(t) = \frac{d}{2}(\cos(\omega_1 t) - \cos(\omega_2 t))$$

Consider trigonometric identities, denote

$$\omega_+ = \frac{1}{2}(\omega_1 + \omega_2), \omega_- = \frac{1}{2}(\omega_1 - \omega_2)$$

we get

$$x_1(t) = d\cos(\omega_+ t)\cos(\omega_- t), x_2(t) = d\sin(\omega_+ t)\sin(\omega_- t)$$

3.5 The Beaded String

On trigonometry,

$$M\ddot{u_n} = -T(sin\psi + sin\phi)$$

The small displacements can be approximated by

$$sin\psi = \frac{u_n - u_{n-1}}{a}, sin\phi = \frac{u_n - u_{n+1}}{a}$$

Thus

$$\ddot{u_n} = \frac{T}{Ma}(u_{n-1} + u_{n+1} - 2u_n)$$

Boundary conditions demands $u_0 = u_{N+1} = 0$.

3.5.1 Normal Modes

$$u_n = A_n e^{i\omega t}$$

Substitute it into the polynomial gives

$$\omega^2 A_n = \frac{T}{Ma}(-A_{n-1} - A_{n+1} + 2A_n)$$

3.5.2 Infinite Chain

Infinite Chain indicates a translational symmetry, as we can pick any random lattice in the chain and observe the same pattern. Therefore naturally, displacements are known as

$$u_n = (\alpha e^{in\theta} + \beta e^{-in\theta})e^{i\omega t}, \omega^2 = \frac{4T}{Ma}sin^2(\frac{\theta}{2})$$

This is the classical analogy to Bloch Waves, and it's obvious that u_n is in the pattern of a superposition of plane wave (both leftward and rightward, the first term) and time-propagation term $e^{i\omega t}$.

3.5.3 Finite Chain

θ is restricted to certain values. ω is therefore fixed too, as a consequence. Therefore,

$$\sin((N+1)\theta) = 0, \theta = \frac{m\pi}{N+1}, m \in \mathbb{Z}, \omega = 2\sqrt{\frac{T}{Ma}}sin(\frac{m\pi}{2(N+1)})$$

4 Rotating Coordinates

For a vector $\boldsymbol{A}$ rotating about the origin in angular velocity $\boldsymbol{\omega}$,

$$\frac{d\boldsymbol{A}}{dt} = \boldsymbol{\omega} \times \boldsymbol{A}$$

This applies to all rotating coordinates. Suppose $\mathcal{O}$ is a stationary frame and $\mathcal{O}'$ is a rotating frame, suppose $\boldsymbol{a} = a_i\hat{i}+a_j\hat{j}+a_k\hat{k}$, then $\boldsymbol{a} = a_{i'}\hat{i}'+a_{j'}\hat{j}'+a_{k'}\hat{k}'$. Thus,

$$\frac{da}{dt} = \frac{da_O}{dt} = \frac{da_{O\prime}}{dt} + \omega \times \boldsymbol{a}$$

4.1 Newton's 2nd Law in rotating frames

Denote the time derivative in frame $\mathcal{O}'$ as $\frac{D}{Dt}$. Then the angular acceleration is

$$\frac{d^2\boldsymbol{r}}{dt^2} = \frac{d}{dt}(\frac{D\boldsymbol{r}}{Dt} + \boldsymbol{\omega} \times \boldsymbol{r}) = \frac{D^2\boldsymbol{r}}{Dt^2} + 2\boldsymbol{\omega} \times \frac{D\boldsymbol{r}}{Dt} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \boldsymbol{r})$$

As we happen to be in frame $\mathcal{O}'$, but N2L, $\boldsymbol{F_{total}} = m\frac{d^2\boldsymbol{r}}{dt^2}$is applied in stationary frame $\mathcal{O}$:

$$m\frac{D^2\boldsymbol{r}}{Dt^2} = \boldsymbol{F_{total}} - 2m\boldsymbol{\omega} \times \frac{D\boldsymbol{r}}{Dt} - m\boldsymbol{\omega} \times (\boldsymbol{\omega} \times \boldsymbol{r})$$

The first term is known as Coriolis Force:

$$\boldsymbol{F_C} = -2m\omega \times \frac{Dr}{Dt}$$

and the second term is combined with g to yield apparent gravity.

4.2 apparent gravity

As the norm of $\boldsymbol{\omega} \times (\boldsymbol{\omega} \times \boldsymbol{r})$ is fixed as $\omega^2 Rcos\lambda$, in which λ is the latitude of the object, the apparent gravity $g^*$is defined by vector extraction

$${g^*}^2 = g^2 + (\omega^2 Rcos\lambda)^2 - 2g\omega^2 Rcos^2\lambda$$

with each λ maps a deflection angle α that satisfies the sine rule

$$\frac{sin\alpha}{\omega^2 Rcos\lambda} = \frac{sin\lambda}{g^*}$$

That is to say,

$$\alpha \approx \frac{\omega^2 R}{2g} sin2\lambda$$

with maximum at latitude 45° with $\alpha \approx 0.1°$.

4.3 Coriolis Force

Pick a random point In the Earth frame(that is, $\mathcal{O}'$ frame), pick $\hat{z}$ antiparallel with the apparent gravity, and $\hat{x}$ eastward, $\hat{y} = \hat{z} \times \hat{x}$. In this coordinate system, at latitude λ ,

$$\boldsymbol{\omega} = \omega \begin{pmatrix} 0 \\ cos\lambda \\ sin\lambda \end{pmatrix}$$

As

$$\frac{D\boldsymbol{r}}{Dt} = \begin{pmatrix} \frac{Dx}{Dt} \\ \frac{Dy}{Dt} \\ \frac{Dz}{Dt} \end{pmatrix}$$

Do the cross product and we'll yield the N2L on x,y,z directions. in z directions, we need to add in the apparent gravity term.

4.4 Coriolis Force on Free-Fall

Coriolis Force creates a detour on xy plane:

$$\frac{D^2\boldsymbol{r}}{Dt} = \boldsymbol{g^*} - 2\omega \times \frac{D\boldsymbol{r}}{Dt}$$

That is

$$\frac{D\boldsymbol{r}}{Dt} = \boldsymbol{v} + \boldsymbol{g}t - 2\omega \times (\boldsymbol{x} - \boldsymbol{x_0})$$

as we only care about 1st order ω, we can substitute the 0th order solution $x = x_0 + \boldsymbol{v}t + \frac{\boldsymbol{g}t^2}{2}$. Thus,

$$\frac{Dx}{Dt} = \boldsymbol{v} + \boldsymbol{g}t - 2\omega \times (\boldsymbol{v}t + \frac{1}{2}gt^2)$$

Take 2 examples:

4.4.1 Dropping mass from a tower

$$\boldsymbol{v} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \boldsymbol{x_0} = \begin{pmatrix} 0 \\ 0 \\ h \end{pmatrix}$$

as

$$\omega \times \boldsymbol{g} = -\omega g cos\lambda \hat{x}$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ h \end{pmatrix} - \frac{gt^2}{2} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} + \frac{\omega g t^3}{3} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

The resultant deflection in x direction is

$$x(z=0) = \frac{1}{3}\omega \cos\lambda \sqrt{(\frac{8h^3}{g})}$$

4.4.2 Projectile Motion

$$\boldsymbol{v} = \frac{1}{\sqrt{2}}\begin{pmatrix}0\\1\\1\end{pmatrix}, \boldsymbol{x_0} = \begin{pmatrix}0\\0\\0\end{pmatrix}$$

This time

$$\omega \times \boldsymbol{v} = \frac{\omega v}{\sqrt{2}}(\cos\lambda - \sin\lambda)\hat{x}$$

Therefore,

$$\begin{pmatrix}x\\y\\z\end{pmatrix} = \frac{vt}{\sqrt{2}}\begin{pmatrix}0\\1\\1\end{pmatrix} - \frac{gt^2}{2}\begin{pmatrix}0\\0\\1\end{pmatrix} - \frac{\omega v t^2}{\sqrt{2}}(\cos\lambda - \sin\lambda)\begin{pmatrix}1\\0\\0\end{pmatrix} + \frac{\omega g t^3}{3}\cos\lambda\begin{pmatrix}1\\0\\0\end{pmatrix}$$

The resultant x displacement is

$$x_f = \frac{\sqrt{2}\omega v^3}{3g^2}(3\sin\lambda - \cos\lambda)$$

Do notice that the sign of x_f is determined by the trigonometric terms:

$$3\sin\lambda - \cos\lambda = 0, \lambda \approx 18.4°$$

If $\lambda < 18.4°$ the mass is deflected to the West, and if $\lambda \in (18.4°, 45°)$, the mass will switch sign in the air (from $\mathcal{O}'$ frame.), but the final displacement is eastward. If $\lambda > 45°$, the displacement is westward.

4.5 Focault Pendulum

$$\frac{D^2x}{Dt^2} = -{\omega_0}^2 x + 2\omega\sin\lambda\frac{Dy}{Dt}, \frac{D^2y}{Dt^2} = -{\omega_0}^2 y - 2\omega\sin\lambda\frac{Dx}{Dt}$$

.

Now if we denote $\alpha = x + iy$, the 2 equations are bounded as:

$$\frac{D^2\alpha}{Dt^2} + {\omega_0}^2\alpha + 2i\omega\sin\lambda\frac{D}{Dt} = 0, \omega_0 = \sqrt{\frac{g}{l}}$$

Characteristic Equation yields

$$\Omega = -\omega\sin\lambda \pm \sqrt{{\omega_0}^2 + \omega^2\sin^2\lambda} \approx -\omega\sin\lambda \pm \omega_0$$

Then the general solution is

$$\alpha = (Ae^{i\omega_0 t} + Be^{-i\omega_0 t})e^{-i\omega\sin\lambda t}$$

which can also be written in trigonometric forms.

5 Particle systems and COM frame

5.1 Conservations

Total momentum is conserved if the system subjects to no external forces; Total angular momentum is conserved if the system subjects to no external torques. This is proven by switching into COM frame, and divide F and τ into internal and external with subscripts. According to N3L, all the internal forces are cancelled off, leaving $F_{CM}^{ext} = \dot{\boldsymbol{p}}$ and $\tau_{CM}^{Ext} = \dot{\boldsymbol{L}}$. If the LHS of equation is 0, the conservancy is proven.

5.2 Center of Mass Frame

COM frame is thus defined as

$$x_{CM} = \frac{\Sigma_i m_i \boldsymbol{r}_i}{\Sigma_i m_i} = \frac{1}{M} \Sigma_i m_i \boldsymbol{r}_i$$

with M defined as total mass. Suppose COM frame is a Galileo Transformation away from stationary frame, $x_{CM} = \rho_i = \boldsymbol{r}_i - \boldsymbol{R}$.
Naturally, the physical picture demands kinetic energy to be

$$T_S = T_{CM} + \Sigma_i \frac{m_i \dot{\rho}_i^2}{2}$$

and total angular momentum is given by

$$\boldsymbol{L} = \boldsymbol{R} \times M\dot{\boldsymbol{R}} + \Sigma_i \rho_i \times M\dot{\boldsymbol{\rho}}_{\boldsymbol{i}}$$

6 Variational Principle of Functionals

$$S[y(x)] = \int_{x_1}^{x_2} L(x, y(x), y'(x))dx$$

is known as a functional. L is decided by the specific situation.
Consider $\delta S = S[y + \delta y] - S[y]$,obviously $\delta S \geq 0$, and the minimum is reached if and only if $\delta S = 0$. Extend the terms, and boundary condition dictates that $\delta y(x_1) = \delta y(x_2) = 0$, this makes

$$\delta S = \int (\frac{d}{dx}\frac{\partial L}{\partial y'} - \frac{\partial L}{\partial y})dx\delta y$$

Setting $\delta S = 0$ yields the Eular-Lagrange Equation directly:

$$\frac{d}{dx}\frac{\partial L}{\partial y'} - \frac{\partial L}{\partial y} = 0$$

This equation has two special cases when:

6.1 L independent of y

$P = \frac{\partial L}{\partial y'}$. is known as the generalised momentum.

6.2 L independent of x

$$H = y' \frac{\partial L}{\partial y'} - L$$

(Legendre Transformation, used for switching to more intuitive variables). This makes

$$\frac{dH}{dx} = y' [\frac{d}{dx} \frac{\partial L}{\partial y'} - \frac{\partial L}{\partial y}] - \frac{\partial L}{\partial x}$$

If L is independent of x, obviously $\dfrac{dH}{dx} = 0$, H is a conserved quantity known as the ***Hamiltonian***. This is used at brachistochrone problem. The key idea here is to take H as a given constant (since it's conserved), and use H to solve for $\frac{dy}{dx}$.

7 Lagrangian Mechanics

7.1 Generalised Coordinates and Degree of Freedom

Usually for a n-Dimensional space there exists n degrees of freedom. Each linear constraint removes 1 DoF; For symmetries, they can be expressed in a group G, and the original system is represented by Q, the new DoF is a quotient group $Q_{res} = Q/G$.
A set of mutually independent quantities used to specify the configuration of the system is known as generalised coordinates, q_i and θ are two common examples. Correspondingly, L has the dimension of energy according to definition (will be introduced soon), thus $\frac{\partial L}{\partial \dot{q}_i}$ is known as generalised momentum, $\frac{\partial L}{\partial q}$ is the generalised force.

7.2 Hamilton's Principle

Hamilton's Principle determines that the $action\ functional$

$$S[\boldsymbol{q}(t)] = \int_{t_1}^{t_2} L(q, \dot{q}, t) dt$$

is an extremum. $\boldsymbol{q}$ is a complete set of generalised coordinates, and L here is referred to as the ***Lagrangian*** with $L = T - U$ as T is the kinetic energy of the system(proven later), and U being the potential energy. Do note that the extremum doesn't necessarily have to be minimum, though they are for most of the times in classical Mechanics.
Now, one can easily recognise that the Eular-Lagrange Equation regarding the action functional is just a generalised N2L: $\frac{d}{dt} \frac{\partial L}{\partial \dot{q}} = \frac{\partial L}{\partial q}$ is in the form of $\dot{p} = \boldsymbol{F}$.

Do note that the generalised force and momentum aren't necessarily in the dimension of force and momentum; For example, torque vs angular momentum can also satisfy the equation.
The Action Functional is also invariant under Galileo Transformation:

$$\int \tilde{L} - L dt = 0$$

This allows

$$\tilde{L} = L + \frac{d}{dt}G(q,t)$$

as

$$\delta(G(q(t_f), t_f) - G(q(t_0), t_0))) = 0$$

due to boundary conditions $\delta(q(t_1)) = \delta(q(t_2)) = 0$, namely $\frac{\partial G}{\partial q}(q,t)\delta q(t_2) - \frac{\partial G}{\partial q}(q,t)\delta q(t_1)$.

7.3 Lagrangian of Free Particle

A Lagrangian is in the form of $L = T - U$ not by definition but by deduction as shown below. For a free particle, the Lagrangian ought not be dependent on location x and time t since there's no preferred location, direction or time. Therefore it must be continuous on $\mathbb{R}^n$ symmetric and Galileo invariant.
This means

$$n > 0, n \in \mathbb{Z}, n \mod 2 = 0$$

whilst

$$\tilde{L} - L = G'(q,t) = \frac{\partial G}{\partial x}\dot{x} + \frac{\partial G}{\partial t}$$

that is at most linear v. Therefore the only choice is

$$L = k\dot{x}^2$$

By definition and newtonian mechanics, $\frac{\partial L}{\partial \dot{x}} = mv$, thus $k = \frac{m}{2}$, $L = T = \frac{mv^2}{2}$ and m is known as the *mass* of the particle.
Extend this to $U(q) \neq 0$ along $\mathbb{R}^n$. This does not change the first term of the E-L equation but left $\frac{\partial L}{\partial q}$ with a generalised force $\frac{dU}{dq}$, therefore $L = T - U$.
For a system of n particles, usually

$$L = \Sigma_n \frac{1}{2} m\dot{x_n}^2 - U(x_n, \dot{x_n})$$

, in which $U(x_n, \dot{x_n}) = U_1(x_n) + U_2(x_n - x_{n'})$.

7.4 Conservations

A conserved quantity $A(q, \dot{q}, t)$ is a quantity that does not change throughout the motion of the system. According to Noether's Theorem, each conserved quantity corresponds to a symmetry: (Momentum, translation), (Energy, time) and (Angular momentum, Rotation).

7.4.1 Conserved Momentum

According to the discussion in section 4, a conserved momentum corresponds to

$$F_j = \frac{\partial L}{\partial q_j} = 0$$

Consider the motion of a mass only subject to gravitational force on Earth.

$$T = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2 + \dot{z}^2), U = mgz, L = T - U$$

Therefore the momenta are:

$$\frac{\partial L}{\partial x} = \frac{\partial L}{\partial y} = 0, p_x = \frac{\partial L}{\partial \dot{x}} = m\dot{x}, p_y = \frac{\partial L}{\partial \dot{y}} = m\dot{y}$$

Furthermore we get EOM:

$$x_t = x_0 + \frac{p_x}{m}t$$

same with y. In z direction,

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{z}} = \frac{\partial L}{\partial z}$$

This reduces to N2L:

$$m\ddot{z} = -mg$$

and leads to the result.
Consider another example in a cylindrical field:

$$U(x, y) = U(\rho), \rho = \sqrt{x^2 + y^2}$$

In polar coordinates (ρ, ϕ) the Lagrangian is therefore written as

$$L = \frac{1}{2}(\dot{\rho}^2 + \rho\dot{\phi}^2) - U(\rho)$$

In this case,

$$\frac{\partial L}{\partial \dot{\phi}} = m\rho^2\dot{\phi}$$

7.4.2 Conserved Energy

Which is just Hamiltonian. Recall the Legendre Transformation:

$$H(q, \dot{q}, t) = \sum_i p_i\dot{q}_i - L(q, \dot{q}, t)$$

Take total time derivative:

$$\frac{dH}{dt} = \sum_{j=1}^{n}(\dot{p}_j\dot{q}_j + p_j\ddot{q}_j - \frac{\partial L}{\partial q_j}\dot{q}_j - \frac{\partial L}{\partial \dot{q}_j}\ddot{q}_j) - \frac{\partial L}{\partial t}$$

This is simply

$$\frac{dH}{dt} = -\frac{\partial L}{\partial t}$$

For $L = \Sigma_{j=1}^{n} \frac{1}{2} m_j \dot{\boldsymbol{r}}_j - U(\boldsymbol{r}_1, ..., \boldsymbol{r}_n)$,the corresponding Hamiltonian would be

$$H = \Sigma_{j=1}^{n} \frac{1}{2} m_j \dot{\boldsymbol{r}}_j + U(\boldsymbol{r}_1, ..., \boldsymbol{r}_n) = T + U$$

But this is not always the case.

7.5 Coordinate systems

Usually we take Cartesian, cylindrical or spherical polar coordinates, and the corresponding T will be given as follows.

7.5.1 Cartesian Coordinates

For a simple mass m,

$$T = \frac{1}{2} m(\dot{x}^2 + \dot{y}^2 + \dot{z}^2)$$

If it's confined to a plane,

$$T = \frac{1}{2} m(\dot{x}^2 + \dot{y}^2)$$

7.5.2 Cylindrical coordinates

$$x = \rho \cos \phi, y = \rho \sin \phi$$

This leads to

$$\dot{x} = \dot{\rho} \cos \phi - \rho \dot{\phi} \sin \phi, \dot{y} = \dot{\rho} \sin \phi + \rho \dot{\phi} \cos \phi$$

The kinetic energy is therefore

$$T = \frac{1}{2} m(\dot{\rho}^2 + \rho^2 \dot{\phi}^2 + \dot{z}^2)$$

If the motion is confined to a plane, remove $\dot{z}^2$.

7.5.3 Spherical Coorinates

This is worth memorising:

$$x = r \sin \theta \cos \phi, y = r \sin \theta \sin \phi, z = r \cos \theta$$

Therefore

$$\begin{aligned}
\dot{x} &= \dot{r} \sin \theta \cos \phi + r \dot{\theta} \cos \theta \cos \phi - r \dot{\phi} \sin \theta \sin \phi \\
\dot{y} &= \dot{r} \sin \theta \sin \phi + r \dot{\theta} \cos \theta \sin \phi + r \dot{\phi} \sin \theta \cos \phi \\
\dot{z} &= \dot{r} \cos \theta - r \dot{\theta} \sin \theta
\end{aligned}$$

This makes the kinetic energy $T = \frac{1}{2} m(\dot{r}^2 + r^2 \dot{\theta}^2 + r^2 \dot{\phi}^2 \sin^2 \theta)$

7.6 Procedure

Step 1. Choose a convenient set of generalised coordinates. $q_1, ..., q_n$.
Step 2. Find $T(q, \dot{q}, t)$, $U(q, \dot{q}, t)$ and the Lagrangian $L(q, \dot{q}, t) = T - U$.
Step 3. Find the generalised momenta $p_j = \frac{\partial L}{\partial \dot{q}_j}$ and the generalised forces $F_j = \frac{\partial L}{\partial q_j}$. Step 4. Use E-L equations to obtain EOM, and identify conserved quantities.

8 Hamiltonian Mechanics

8.1 Noether's Theorem

Noether's Theorem states that to each continous symmetry of a system there is a corresponding Noether charge.
Consider $q_\sigma \to \tilde{q_\sigma}(q, \zeta)$ in which ζ is a continuous parameter. $\zeta = 0$ represents the identity. Perform this on the Lagrangian $L(\tilde{q}, \dot{\tilde{q}}, t)$:

$$\frac{d}{d\zeta}\bigg|_{\zeta=0} L(\tilde{q}, \dot{\tilde{q}}, t) = 0$$

That is

$$\sum_\sigma \frac{\partial L}{\partial \tilde{q_\sigma}} \frac{\partial \tilde{q_\sigma}}{\partial \zeta}\bigg|_{\zeta=0} + \frac{\partial L}{\partial \dot{\tilde{q_\sigma}}} \frac{\partial \dot{\tilde{q_\sigma}}}{\partial \zeta}\bigg|_{\zeta=0} = 0$$

$$\sum_\sigma \frac{d}{dt} \frac{\partial L}{\partial \dot{\tilde{q_\sigma}}} \frac{\partial \tilde{q_\sigma}}{\partial \zeta}\bigg|_{\zeta=0} + \frac{\partial L}{\partial \dot{\tilde{q_\sigma}}} \frac{d}{dt} \frac{\partial \tilde{q_\sigma}}{\partial \zeta}\bigg|_{\zeta=0} = 0$$

The conserved Noether charge is therefore:

$$\Lambda = \sum_\sigma \left(\frac{\partial L}{\partial \dot{\tilde{q_\sigma}}} \frac{\partial \tilde{q_\sigma}}{\partial \zeta}\right)\bigg|_{\zeta=0}$$

Do note that it's more applicable than you thought. Take 2 examples:

- Example 1:

$$L = \frac{1}{2} m(\dot{r}^2 + r^2 \dot{\phi}^2) - U(r)$$

  The transformation is:

$$r(\zeta) = r, \phi(\zeta) = \phi + \zeta$$

  This leaves $\partial_\zeta r = 0$ and $\partial_\zeta \phi = 1$. Putting in Λ we get the Noether Charge is:

$$\Lambda = mr^2 \dot{\phi}$$

- Example 2:

$$L = \frac{1}{2} m(\dot{\rho}^2 + \rho^2 \dot{\phi}^2 + z^2) - V(\rho, a\phi + z)$$

The transformation is defined as:

$$\tilde{\rho}(\zeta) = \rho, \tilde{\phi}(\zeta) = \phi + \zeta, \tilde{z}(\zeta) = z - \zeta a$$

The result is

$$\Lambda = m\rho^2\dot{\phi} - ma\dot{z}$$

To test the time dependency:

$$\frac{d}{dt}\Lambda = 0$$

8.2 Hamilton's Picture

The Lagrange equation of $q_i, i \in [1, n]$ is

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{q}_i} - \frac{\partial L}{\partial q_i} = 0$$

This leaves us with n second order ODEs demanding $2n$ initial conditions to be solved.
If we know q_i and $\dot{q}_i$, Hamilton's plan is to eliminate $\dot{q}_i$ by p_i, creating a pair of (q_i, p_i), known as a *state* in the *phase space* where paths can never cross each other. The evolution of the system is therefore defined as a $flow$ in the phase space.
Hamilton's Equations can be derived from

$$H(\boldsymbol{q}, \boldsymbol{p}, t) = \sum_i p_i\dot{q}_i - L(\boldsymbol{q}, \dot{\boldsymbol{q}}, t)$$

as

$$\dot{q}_i = \frac{\partial H}{\partial p_i}, \dot{p}_i = -\frac{\partial H}{\partial q_i}$$

By doing such, we reduced n 2^{nd} order ODEs into $2n$ first order ODEs.

8.3 Hamiltonian and Energy

It's highly important to remember that while Lagrangian is always defined as $T - U$, the Hamiltonian isn't always equivalent to the system's energy.
Consider a general form of kinetic energy:

$$T = T^{(2)}_{\sigma\sigma'}(q,t)\dot{q}_\sigma\dot{q}_{\sigma'} + T^{(1)}_\sigma(q,t)\dot{q_\sigma} + T^{(0)}(q,t)$$

while T is symmetric: $T^{(n)}_{\sigma\sigma'} = T^{(n)}_{\sigma'\sigma}$. Assume the potential is in form of

$$U = U_1 + U_0 = U^{(1)}_\sigma(q,t)\dot{q_\sigma} + U^{(0)}_\sigma(q,t)$$

Then perform $L = T - U$ and gain

$$p_\sigma = \frac{\partial L}{\partial \dot{q_\sigma}} = T^{(2)}\dot{q_{\sigma'}} + T^{(1)} - U^{(1)}, \dot{q}_\sigma = (T^{(2)}_{\sigma\sigma'})^{-1}(p_{\sigma'} - T^{(1)} + U^{(1)})$$

This leaves the Hamiltonian:

$$H = p_\sigma \dot{q_\sigma} - L$$

$$H = \frac{1}{2}(T^{(2)}_{\sigma\sigma'})^{-1}(p_\sigma - T^{(1)}_\sigma + U^{(1)}_\sigma)(p_{\sigma'} - T^{(1)}_{\sigma'} + U^{(1)}_{\sigma'}) - T_0 + U_0$$

Therefore, $H = T+U$ only if T_0, T_1 and U_1 all vanish. In other words, only if the kinetic energy is a homogeneous quadratic function of the generalised velocity, and the potential is velocity-independent. Otherwise, $H \neq T + U$.

8.4 Canonical Transformations, Generators and Poisson Brackets

Refer to Quantum Mechanics Notes (Section 1, "Dirac Rule") for the bridge between AM and QM. As indicated before, Hamiltonian Mechanics describes the system as a state (q_i, p_i) in phase space. Hamiltonian Mechanics studies how the state evolves in time, that is

$$(q_i(0), p_i(0)) \rightarrow (q_i(t), p_i(t))$$

This is known as the "Hamiltonian Flow". Since a state is expressed as $(\boldsymbol{q}, \boldsymbol{p})$ as $\boldsymbol{q} = (q_1, q_2, ..., q_n), \boldsymbol{p} = (p_1, p_2, ..., p_n)$, the Hamiltonian flow always happens in a 2n-Dimensional space: sympletic Geometry.
Recall the principle of least action. In Hamiltonian Mechanics, this is translated as Hamiltonian flow always flows in a contour that's perpendicular to ∇H: ∇H is the steepest, therefore the direction always perpendicular to it is $J\nabla H$. Now consider only one point in the phase space($n = 1$), J is known as sympletic matrix:

$$J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

The contour in Hamiltonian Flow is

$$X_H = J\nabla H = \begin{pmatrix} \dot{q_i} \\ \dot{p_i} \end{pmatrix} = \begin{pmatrix} \partial_p H \\ -\partial_q H \end{pmatrix}$$

Therefore, consider an observable A, it is obviously determined by the state: $A(q(t), p(t))$. ∇A reveals the gradient of observable A in the phase space:

$$\nabla A = (\frac{\partial A}{\partial q}, \frac{\partial A}{\partial p})$$

This is known as the amplitude of Hamiltonian Flow, and the actual flow direction and speed are given by X_H. Therefore, the dot product of ∇A and X_H can be abbreviated into Poisson Brackets:

$$\nabla A \cdot X_H = \frac{\partial A}{\partial q}\frac{\partial H}{\partial p} - \frac{\partial A}{\partial p}\frac{\partial H}{\partial q} = \{A, H\}$$

One way to imagine this is since X_H is a 2-D vector field (w.r.t. a single state), picture $A(q,p)$ as a mountain and X_H gives out a special direction, a path on the mountain. Luckily according to definition(the Legendre transformation, really), the direction of Hamiltonian flow is the direction of time!
$\{A, H\}$ therefore tells you how A changes along the Hamiltonian flow H. In QM, the state space is generalised into Hilbert space, and PB changes to commutators, but the main idea remains the same: a path on the mountain.
It goes beyond saying that Hamiltonian is no different from other functions, therefore you can find any function in the phase space: $G(q,p)$ and an observable $A(q,p)$ and still create the Poisson bracket $\{A, G\}$, but then it no longer resembles time translation but something else according to Noether's Theorem.
A canonical coordinate is a set of (q,p) that satisfies

$$\{q_i, p_j\} = \delta_{ij}, \{q_i, q_j\} = 0, \{p_i, p_j\} = 0$$

A canonical flow is thus

$$\frac{dA}{d\lambda} = \{A, G\} + \frac{\partial A}{\partial \lambda}$$

if $G = H$,then $\lambda = t$. By satisfying $\{A, G\} = 0$, A does not change along the G flow, and therefore forms a classical analogue of compatible operators in QM.

9 Central Forces and Planetary Motion

9.1 Two-body central forces problem

9.1.1 COM Coordinates and its motion

Lagrangian of such system is:

$$L = T - U = \frac{1}{2}m_1\boldsymbol{r}_1^2 + \frac{1}{2}m_2\boldsymbol{r}_2^2 - U(|r_1 - r_2|)$$

If we recall the definition of COM, we take the COM coordinate:

$$\boldsymbol{R} = \frac{m_1\boldsymbol{r}_1 + m_2\boldsymbol{r}_2}{m_1 + m_2}, \boldsymbol{r} = \boldsymbol{r}_1 - \boldsymbol{r}_2$$

$$\boldsymbol{r}_1 = \boldsymbol{R} + \frac{m_2}{m_1 + m_2}\boldsymbol{r}, \boldsymbol{r}_2 = \boldsymbol{R} - \frac{m_1}{m_1 + m_2}\boldsymbol{r}$$

Then the Lagrangian can be re-expressed as

$$L = \frac{1}{2}M\dot{\boldsymbol{R}}^2 + \frac{1}{2}\mu\dot{\boldsymbol{r}}^2 - U(\boldsymbol{r})$$

in which M is the total mass of the system and μ is the reduced mass:

$$M = m_1 + m_2; \mu = \frac{m_1 m_2}{m_1 + m_2}$$

This enabled us to study the motion of COM without considering relative motions. Do note that

$$\boldsymbol{P} = \frac{\partial L}{\partial \dot{\boldsymbol{R}}} = \text{const}$$

So the total momentum is conserved: $\boldsymbol{R}(t) = R(0) + \dot{\boldsymbol{R}}(0)t$.

9.1.2 Relative motion

- Conservation of Angular momentum:
  Adopt to polar coordinates (r, ϕ) makes the Lagrangian

$$L = \frac{1}{2}\mu\dot{r}^2 + \frac{1}{2}\mu r^2\dot{\phi}^2 - U(r)$$

  As the angular momentum writes $\boldsymbol{l} = \boldsymbol{r} \times \boldsymbol{p} = \mu\boldsymbol{r} \times \dot{\boldsymbol{r}}$, $\boldsymbol{l} = \frac{\partial L}{\partial \dot{\phi}} = 2\mu A$, in which $dA = \frac{1}{2}r^2 d\phi$ is the object desribed in Keplar's 2nd Law.

- Conservation of Energy:

$$\frac{d}{dt}(\frac{\partial L}{\partial \dot{\boldsymbol{r}}}) - \frac{\partial L}{\partial \boldsymbol{r}} = 0 \rightarrow \mu\ddot{\boldsymbol{r}} + \frac{dU}{d\boldsymbol{r}} = 0$$

  Taking dot product with $\dot{\boldsymbol{r}}$ gets us

$$\frac{d}{dt}(\frac{1}{2}\mu\dot{\boldsymbol{r}}^2 + U(r)) = 0$$

  That indicated energy conservation.

- Effective potential:
  To treat the term angular kinetic energy as potential terms also:

$$L = \frac{1}{2}\mu\dot{r}^2 + \frac{1}{2}\mu r^2\dot{\phi}^2 - U(r); E = \frac{1}{2}\mu\dot{r}^2 + U_{eff}(r)$$

  in which

$$U_{eff}(r) = \frac{1}{2}\mu r^2\dot{\phi}^2 + U(r) = \frac{l^2}{2\mu r^2} + U(r)$$

  U_{eff} has a local minimum indicated a stable orbit. Otherwise there are no stable orbit.

- Integrations of EoM:
  Energy is now $E = \frac{1}{2}\mu\dot{r}^2 + \frac{l^2}{2\mu r^2} + U(r)$. As $\frac{dE}{dt} = 0$, we get the EoM

$$\mu\ddot{r} = -\frac{dU_{eff}}{dr}$$

  Integrate it twice to get $r(t)$.

- Geometric Orbit Equation:
  If we want to obtain this equation, we must eliminate the time-dependence,that is to get $r(\phi)$ from $\frac{dr}{d\phi}$. Now consider $l = \mu r^2\dot{\phi}$, we can rewrite it in the form of

$$\frac{d}{dt} = \frac{l}{\mu r^2}\frac{d}{d\phi}$$

  After chain rule:

$$\frac{dr}{d\phi} = \frac{dr}{dt}(\frac{d\phi}{dt})^{-1}$$

So we'll have to notice:

$$\frac{d^2r}{d\phi^2}-\frac{2}{r}(\frac{dr}{d\phi})^2=\frac{d}{d\phi}(\frac{\mu r^2}{l}\dot{r})-\frac{2}{r}(\frac{\mu r^2}{l}\dot{r})^2=r+\frac{\mu r^4}{l^2}F(r)$$

Where

$$F(\boldsymbol{r})=-\frac{dU_{eff}}{dr}$$

This form is later used to simplify the equation. If we adopt to $s=\frac{1}{r}$, then the equation reduces to:

$$\frac{ds}{d\phi}=-\frac{1}{r^2}\frac{dr}{d\phi},\frac{d^2s}{d\phi^2}=\frac{2}{r^3}(\frac{dr}{d\phi})^2-\frac{1}{r^2}\frac{d^2r}{d\phi^2}$$

This makes

$$\frac{d^2s}{d\phi^2}+s=-\frac{\mu}{l^2s^2}F(\frac{1}{s})$$

9.2 Almost Circular Orbits:Perturbation

Everything about circular orbits is concluded–They must satisfy the condition

$$F(r_0)=-\frac{l^2}{\mu r_0^3}=-\frac{dU}{dr}\bigg|_{r=r_0}$$

Perturbation is known as modifying r_0 to $r_0+\eta(t)$ where $|\frac{\eta(t)}{r_0}|\ll 1$, therefore allowing us to expand

$$\mu\ddot{\eta}=\frac{dU_{eff}}{dr}\bigg|_{r=r_0+\eta}=-(\frac{dU}{dr}+\eta^2\frac{d^2U}{dr^2})_{r=r_0}$$

the first term is 0 at r_0 since $\ddot{r_0}=0\to\ddot{r}=\ddot{\eta}$. The equation is thus

$$\frac{d^2\eta}{dt^2}=-\omega^2\eta$$

in which

$$\omega^2=\frac{1}{\mu}\frac{d^2U_{eff}}{dr^2}\bigg|_{r=r_0}$$

If $\omega^2>0$ the orbit is stable, and will oscillate in the corresponding trigonometric terms; otherwise the orbit is unstable and the perturbation will grow exponentially. Shape of the orbit is given by:

$$\frac{d^2\eta}{dt^2}=\eta+\frac{\mu r_0^4}{l^2}\eta\frac{dF}{dr}\bigg|_{r=r_0}+\frac{4\mu r_0^3\eta}{l^2}F(r_0)$$

That is,

$$\frac{d^2\eta}{d\phi^2}=-\beta^2\eta$$

Where

$$\beta^2 = (3 - \frac{\mu r_0^4}{l^2}\frac{dF}{dr}\bigg|_{r=r_0})$$

if $\beta^2 > 0$ the solution would be:

$$\eta(\phi) = \eta_0 \cos(\beta(\phi - \delta_0))$$

At which η_0 and δ_0 are given by initial conditions. Therefore, η oscillates during the orbit; This phenomenon is known as precession.
The points at which the radius is maximum are known as the apoapsis:

$$\phi_n = \delta_0 + \frac{2n\pi}{\beta}$$

Correspondingly radius minimum are known as the periapsis points, with phase shift π compared to apoapsis.The orbit advances by the amount of:

$$\Delta\phi = 2\pi(\frac{1}{\beta} - 1)$$

If $\beta > 1$, the apoapsis/periapsis points advance, and the orbit rotates $\phi' < 2\pi$ due to precession, vice versa; If $\beta = \frac{p}{q}$, the orbit is closed, that is, the orbit returns to the initial state after q full rotations.

9.3 Precession, in an exact model

Notice that the equation

$$r(\phi) = \frac{r_0}{1 - \epsilon\cos(\beta\phi)}, \epsilon \in [0, 1]$$

is an exact solution to the EoM above, proven as follows. As now

$$s = s_0[1 - \epsilon\cos(\beta\phi)]$$

Substitute it into the EoM:

$$\frac{d^2 s}{d\phi^2} + s = -\frac{\mu}{l^2 s^2}F(\frac{1}{s})$$

We'll get

$$-\frac{\mu}{l^2 s^2}F(\frac{1}{s}) = (1 - \beta^2)s + \beta^2 s_0$$

This leaves $F(r)$ in the form of (recall $s = \frac{1}{r}$):

$$F(r) = -\frac{k}{r^2} - \frac{C}{r^3}, k = \beta^2 s_0 \frac{l^2}{\mu}; C = (\beta^2 - 1)\frac{l^2}{\mu}$$

If $C = 0$ we restored the Gravtitional Law between two objects; We can also say

$$\beta = \sqrt{\frac{\mu C}{l^2} + 1}; r_0 = \frac{\beta^2 l^2}{\mu k} = \frac{l^2}{\mu k} + \frac{C}{k}$$

And the corresponding potential can be given by

$$U(r) = -\int F(r)dr = -\frac{k}{r} + \frac{C}{2r^2} + U_\infty, U_\infty = 0$$

The total energy is therefore given by:

$$E = \frac{1}{2}\mu\dot{r}^2 + U_{eff}(r) = \frac{\mu}{2}(\frac{dr}{d\phi}\dot{\phi})^2 + U_{eff}(r) = \frac{l^2}{2\mu r^4}(\frac{dr}{d\phi})^2 + U_{eff}(r)$$

Recall from definition that

$$\frac{dr}{d\phi} = -\frac{r_0}{1-\epsilon\cos(\beta\phi)}\epsilon\sin(\beta\phi)$$

As the energy is conserved anyway, we are allowed to choose an arbitrary point on the orbit to calculate the energy. For the sake of simplicity we'll say $\phi = \frac{\pi}{2\beta}$ where $r = r_0$. The Energy Equation is therefore

$$E = \frac{l^2}{2\mu r_0^4}(r_0\epsilon\beta^2) + \frac{l^2}{2\mu r_0^2}(l^2 + \mu C) - \frac{k}{r_0}$$

Reduces to:

$$E = \frac{\mu k^2}{2(l^2+\mu C)}(\epsilon^2 - 1)$$

9.4 Kepler's problem

In Kepler's problem, $U(r) = -\frac{k}{r}$. This makes $F(r) = -\frac{dU(r)}{dr} = -\frac{k}{r^2}$. This determines

9.4.1 Shape of Geometrical Orbit

Put F(r) in the equation

$$\frac{d^2s}{d\phi^2} + s = -\frac{\mu}{l^2s^2}F = \frac{\mu k}{l^2}$$

The most general solution is

$$s(\phi) = s_0 - C\cos(\phi - \phi_0)$$

Returning to the r-form:

$$r(\phi) = \frac{r_0}{1-\epsilon\cos(\phi-\phi_0)}, r_0 = \frac{l^2}{\mu k}, \epsilon = Cr_0$$

9.4.2 Conic sections

No need to say, the equation above can be divided into:

- Circle: $\epsilon = 0$; $E = -\frac{\mu k^2}{2l^2}$ and $r = \frac{l^2}{\mu k}$.
- Ellipse: $\epsilon \in (0, 1)$, $E \in (-\frac{\mu k^2}{2l^2}, 0)$, r is characterised by semimajor axis $a = -\frac{k}{2E}$ and semiminor axis $b = a\sqrt{1 - \epsilon^2}$.
- Parabola: $\epsilon = 1$,$E = 0$, the first unbound solution.
- Hyperbola: $\epsilon > 1$ thus $E > 0$, will not be bounded by the central mass.

9.4.3 Period and Escape velocity of Kepler Orbits

Recall from high school that

$$\frac{a^3}{T^2} = \frac{GM}{4\pi^2}$$

And the escape velocity are given by:

$$v_{esc} = \sqrt{\frac{2G(m_1 + m_2)}{R}}$$

10 Rigid Body Rotation

10.1 Rigid Body Degrees of Freedom

A rigid body is a collection of particles whose relative distances are fixed:

$$|\mathbf{r}_i - \mathbf{r}_j| = \text{constant}.$$

A general rigid body in three dimensions has six degrees of freedom:

$$3 \text{ translational coordinates} + 3 \text{ rotational coordinates}.$$

Usually these are described by the center-of-mass position

$$\mathbf{R} = (X, Y, Z)$$

and three angles specifying the orientation of the body.

10.2 Angular Velocity

Consider a rigid body rotating about a fixed point. For an infinitesimal rotation

$$d\boldsymbol{\phi} = \hat{\mathbf{n}}\, d\phi,$$

the displacement of a point at position $\mathbf{r}$ is

$$d\mathbf{r} = d\boldsymbol{\phi} \times \mathbf{r}.$$

Dividing by dt, we get the velocity field of a rigid body:

$$\boxed{\dot{\mathbf{r}} = \boldsymbol{\omega} \times \mathbf{r}}$$

where

$$\boldsymbol{\omega} = \frac{d\boldsymbol{\phi}}{dt}$$

is the instantaneous angular velocity.

This equation is the starting point of rigid body rotation.

10.3 Rotational Kinetic Energy and the Inertia Tensor

The kinetic energy of a rotating rigid body is

$$T = \frac{1}{2} \sum_i m_i \dot{\mathbf{r}}_i^{\,2}.$$

Using

$$\dot{\mathbf{r}}_i = \boldsymbol{\omega} \times \mathbf{r}_i,$$

we get

$$T = \frac{1}{2} \sum_i m_i (\boldsymbol{\omega} \times \mathbf{r}_i) \cdot (\boldsymbol{\omega} \times \mathbf{r}_i).$$

Using the identity

$$(\mathbf{A} \times \mathbf{B}) \cdot (\mathbf{C} \times \mathbf{D}) = (\mathbf{A} \cdot \mathbf{C})(\mathbf{B} \cdot \mathbf{D}) - (\mathbf{B} \cdot \mathbf{C})(\mathbf{A} \cdot \mathbf{D}),$$

we obtain

$$T = \frac{1}{2} \sum_i m_i \left[\omega^2 r_i^2 - (\boldsymbol{\omega} \cdot \mathbf{r}_i)^2 \right].$$

This can be written in tensor form:

$$\boxed{T = \frac{1}{2} \omega_a I_{ab} \omega_b}$$

where the inertia tensor is

$$\boxed{I_{ab} = \sum_i m_i \left(r_i^2 \delta_{ab} - r_{i,a} r_{i,b} \right)}$$

or, in matrix form,

$$I = \sum_i m_i \begin{pmatrix} y_i^2 + z_i^2 & -x_i y_i & -x_i z_i \\ -x_i y_i & x_i^2 + z_i^2 & -y_i z_i \\ -x_i z_i & -y_i z_i & x_i^2 + y_i^2 \end{pmatrix}.$$

For a continuous body,

$$\sum_i m_i \longrightarrow \int \rho(\mathbf{r})\, d^3 r,$$

so

$$\boxed{I_{ab} = \int \rho(\mathbf{r}) \left(r^2 \delta_{ab} - r_a r_b \right) d^3 r}$$

10.4 Angular Momentum of a Rigid Body

The angular momentum is

$$\mathbf{L} = \sum_i m_i \mathbf{r}_i \times \dot{\mathbf{r}}_i.$$

Using

$$\dot{\mathbf{r}}_i = \boldsymbol{\omega} \times \mathbf{r}_i,$$

we get

$$\mathbf{L} = \sum_i m_i \mathbf{r}_i \times (\boldsymbol{\omega} \times \mathbf{r}_i).$$

Using

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = (\mathbf{A} \cdot \mathbf{C})\mathbf{B} - (\mathbf{A} \cdot \mathbf{B})\mathbf{C},$$

this becomes

$$\mathbf{L} = \sum_i m_i \left[r_i^2 \boldsymbol{\omega} - \mathbf{r}_i(\mathbf{r}_i \cdot \boldsymbol{\omega}) \right].$$

Therefore, in components,

$$\boxed{L_a = I_{ab}\omega_b}$$

or

$$\boxed{\mathbf{L} = I\boldsymbol{\omega}.}$$

Unlike the scalar moment-of-inertia case, $\mathbf{L}$ is not generally parallel to $\boldsymbol{\omega}$.

10.5 Coordinate Transformations

Under a rotation of coordinate axes,

$$\mathbf{e}'_a = R_{ab}\mathbf{e}_b,$$

where R is an orthogonal matrix:

$$R^T R = RR^T = 1.$$

Vector components transform as

$$A'_a = R_{ab}A_b.$$

The inertia tensor transforms as

$$\boxed{I' = RIR^T.}$$

The kinetic energy remains invariant:

$$T' = \frac{1}{2}\boldsymbol{\omega}'^T I' \boldsymbol{\omega}' = \frac{1}{2}\boldsymbol{\omega}^T I \boldsymbol{\omega} = T.$$

10.6 Principal Axes of Inertia

Since the inertia tensor is real and symmetric,

$$I_{ab} = I_{ba},$$

it can be diagonalized by an orthogonal transformation. In the principal-axis frame,

$$\boxed{I = \begin{pmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{pmatrix}}$$

where I_1, I_2, I_3 are the principal moments of inertia.

In this frame,

$$T = \frac{1}{2}(I_1\omega_1^2 + I_2\omega_2^2 + I_3\omega_3^2),$$

and

$$\mathbf{L} = (I_1\omega_1, I_2\omega_2, I_3\omega_3).$$

10.7 Examples of Inertia Tensors

10.7.1 Thin Rod

For a thin rod of length l and mass M, with the z-axis along the rod and origin at its center,

$$I = \begin{pmatrix} I_1 & 0 & 0 \\ 0 & I_1 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

where

$$I_1 = \frac{M}{l}\int_{-l/2}^{l/2} z^2\,dz = \frac{1}{12}Ml^2.$$

Thus

$$\boxed{I = \frac{1}{12}Ml^2 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}.}$$

10.7.2 Thin Disk

For a thin disk of radius R and mass M, with the z-axis perpendicular to the disk,

$$I_1 = I_2 = \frac{1}{4}MR^2, \qquad I_3 = \frac{1}{2}MR^2.$$

Thus

$$\boxed{I = MR^2 \begin{pmatrix} \frac{1}{4} & 0 & 0 \\ 0 & \frac{1}{4} & 0 \\ 0 & 0 & \frac{1}{2} \end{pmatrix}.}$$

10.8 Rigid Body with Translation

If the body is not fixed at a point, separate the motion into center-of-mass motion and rotation about the center of mass:

$$\boxed{T = \frac{1}{2}M\dot{\mathbf{R}}^2 + \frac{1}{2}I_{ab}\omega_a\omega_b.}$$

The total angular momentum is

$$\boxed{\mathbf{L} = M\mathbf{R}\times\dot{\mathbf{R}} + I\boldsymbol{\omega}.}$$

10.9 Parallel Axis Theorem

Suppose the inertia tensor about the center of mass is $I_{ab}(0)$. If the origin is shifted by a vector $\mathbf{d}$, then

$$\boxed{I_{ab}(\mathbf{d}) = I_{ab}(0) + M(d^2\delta_{ab} - d_a d_b).}$$

For a thin rod, shifting the origin from its center to one end gives

$$I = \frac{1}{3}Ml^2\begin{pmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 0\end{pmatrix}.$$

10.10 Euler's Equations

Choose the body-fixed axes to be the principal axes. Then

$$I = \begin{pmatrix}I_1 & 0 & 0\\ 0 & I_2 & 0\\ 0 & 0 & I_3\end{pmatrix},$$

and

$$\mathbf{L} = (I_1\omega_1, I_2\omega_2, I_3\omega_3).$$

The rotational equation of motion is

$$\frac{d\mathbf{L}}{dt} = \mathbf{N}.$$

However, in the rotating body frame, the basis vectors also change:

$$\frac{d\hat{\mathbf{e}}_a}{dt} = \boldsymbol{\omega}\times\hat{\mathbf{e}}_a.$$

Therefore,

$$\frac{d\mathbf{L}}{dt} = \dot{L}_a\hat{\mathbf{e}}_a + L_a\frac{d\hat{\mathbf{e}}_a}{dt}.$$

This gives Euler's equations:

$$\boxed{N_1 = I_1\dot{\omega}_1 + \omega_2\omega_3(I_3 - I_2)}$$

$$\boxed{N_2 = I_2\dot{\omega}_2 + \omega_3\omega_1(I_1 - I_3)}$$

$$\boxed{N_3 = I_3\dot{\omega}_3 + \omega_1\omega_2(I_2 - I_1)}$$

For a free rigid body,

$$\mathbf{N} = 0.$$

10.11 Free Symmetric Top

For a symmetric top,

$$I_1 = I_2 \neq I_3.$$

With no torque, Euler's equations give

$$\dot{\omega}_3 = 0.$$

Thus the spin about the symmetry axis is constant:

$$\omega_3 = \text{constant}.$$

The remaining equations are

$$\dot{\omega}_1 = \left(\frac{I_1 - I_3}{I_1}\right)\omega_3\omega_2,$$

$$\dot{\omega}_2 = \left(\frac{I_3 - I_1}{I_1}\right)\omega_3\omega_1.$$

Define

$$\Omega = \left(\frac{I_3 - I_1}{I_1}\right)\omega_3.$$

Then

$$\dot{\omega}_1 = -\Omega\omega_2, \qquad \dot{\omega}_2 = \Omega\omega_1.$$

The solution is

$$\boxed{\omega_1(t) = \omega_0\cos(\Omega t + \delta),}$$

$$\boxed{\omega_2(t) = \omega_0\sin(\Omega t + \delta).}$$

Thus in the body frame, $\boldsymbol{\omega}$ precesses about the symmetry axis.

10.12 Asymmetric Top and Stability

For a general asymmetric top,

$$I_1 \neq I_2 \neq I_3.$$

Consider rotation mainly about the third principal axis:

$$\boldsymbol{\omega} = \omega_0 \hat{\mathbf{e}}_3 + \delta\boldsymbol{\omega}.$$

Keeping only terms linear in the perturbation gives

$$\begin{aligned} I_1 \dot{\delta\omega}_1 &= \omega_0 \delta\omega_2 (I_2 - I_3), \\ I_2 \dot{\delta\omega}_2 &= \omega_0 \delta\omega_1 (I_3 - I_1), \\ I_3 \dot{\delta\omega}_3 &= 0. \end{aligned}$$

Taking another time derivative gives

$$\boxed{\ddot{\delta\omega}_1 = -\Omega^2 \delta\omega_1,}$$

$$\boxed{\ddot{\delta\omega}_2 = -\Omega^2 \delta\omega_2,}$$

where

$$\boxed{\Omega^2 = \omega_0^2 \frac{(I_3 - I_2)(I_3 - I_1)}{I_1 I_2}.}$$

If $\Omega^2 > 0$, the perturbation oscillates and the rotation is stable.
If $\Omega^2 < 0$, the perturbation grows exponentially and the rotation is unstable.
Therefore:

$$\boxed{\text{rotation about the largest or smallest principal moment is stable;}}$$

$$\boxed{\text{rotation about the intermediate principal moment is unstable.}}$$

10.13 Euler Angles

A general rotation can be parametrized by three Euler angles:

$$(\phi, \theta, \psi).$$

The rotation is decomposed into three steps:

$$R(\phi, \theta, \psi) = R_z(\psi) R_x(\theta) R_z(\phi).$$

The angular velocity in the body frame is

$$\boxed{\boldsymbol{\omega} = (\dot{\phi} \sin\theta \sin\psi + \dot{\theta} \cos\psi)\hat{\mathbf{e}}_1 + (\dot{\phi} \sin\theta \cos\psi - \dot{\theta} \sin\psi)\hat{\mathbf{e}}_2 + (\dot{\psi} + \dot{\phi} \cos\theta)\hat{\mathbf{e}}_3.}$$

Thus,

$$\begin{aligned} \omega_1 &= \dot{\phi} \sin\theta \sin\psi + \dot{\theta} \cos\psi, \\ \omega_2 &= \dot{\phi} \sin\theta \cos\psi - \dot{\theta} \sin\psi, \\ \omega_3 &= \dot{\psi} + \dot{\phi} \cos\theta. \end{aligned}$$

10.14 Free Symmetric Top in the Observer Frame

For a free symmetric top, choose the space-fixed axis $\tilde{\mathbf{e}}_3$ along the conserved angular momentum $\mathbf{L}$.

Then θ is constant:

$$\dot{\theta} = 0.$$

The body-frame precession frequency is

$$\Omega = \dot{\psi} = \left(\frac{I_1 - I_3}{I_1}\right)\omega_3.$$

Using

$$\omega_3 = \dot{\psi} + \dot{\phi}\cos\theta,$$

one obtains

$$\boxed{\dot{\phi} = \frac{I_3\omega_3}{I_1\cos\theta}.}$$

10.15 Heavy Symmetric Top

Consider a symmetric top pinned at a point in a gravitational field. The center of mass is a distance l from the fixed point.

The Lagrangian is

$$L = \frac{1}{2}I_1(\omega_1^2 + \omega_2^2) + \frac{1}{2}I_3\omega_3^2 - Mgl\cos\theta.$$

Using Euler angles,

$$L = \frac{1}{2}I_1\left(\dot{\theta}^2 + \dot{\phi}^2\sin^2\theta\right) + \frac{1}{2}I_3\left(\dot{\psi} + \dot{\phi}\cos\theta\right)^2 - Mgl\cos\theta.$$

Since L does not depend explicitly on ψ, p_ψ is conserved:

$$\boxed{p_\psi = \frac{\partial L}{\partial\dot{\psi}} = I_3(\dot{\psi} + \dot{\phi}\cos\theta) = I_3\omega_3.}$$

Since L does not depend explicitly on ϕ, p_ϕ is also conserved:

$$\boxed{p_\phi = \frac{\partial L}{\partial\dot{\phi}} = I_1\dot{\phi}\sin^2\theta + I_3\omega_3\cos\theta.}$$

Define

$$a = \frac{I_3\omega_3}{I_1}, \qquad b = \frac{p_\phi}{I_1}.$$

Then

$$\boxed{\dot{\phi} = \frac{b - a\cos\theta}{\sin^2\theta}.}$$

Also,

$$\boxed{\dot{\psi} = \omega_3 - \dot{\phi}\cos\theta.}$$

The conserved energy is

$$E = \frac{1}{2}I_1\left(\dot{\theta}^2 + \dot{\phi}^2\sin^2\theta\right) + \frac{1}{2}I_3\omega_3^2 + Mgl\cos\theta.$$

Define the reduced energy

$$E' = E - \frac{1}{2}I_3\omega_3^2.$$

Then the motion in θ becomes one-dimensional:

$$\boxed{E' = \frac{1}{2}I_1\dot{\theta}^2 + U_{\text{eff}}(\theta)}$$

where

$$\boxed{U_{\text{eff}}(\theta) = \frac{I_1(b - a\cos\theta)^2}{2\sin^2\theta} + Mgl\cos\theta.}$$

Using

$$u = \cos\theta,$$

and defining

$$\alpha = \frac{2E'}{I_1}, \qquad \beta = \frac{2Mgl}{I_1},$$

the equation becomes

$$\boxed{\dot{u}^2 = f(u)}$$

where

$$\boxed{f(u) = (\alpha - \beta u)(1 - u^2) - (b - au)^2.}$$

The physical motion is confined to the region where

$$-1 \le u \le 1, \qquad f(u) \ge 0.$$

Thus the heavy symmetric top can be understood as one-dimensional motion in an effective potential.

10.16 Conceptual Summary

The chapter follows the chain:

$$\boxed{\dot{\mathbf{r}} = \boldsymbol{\omega} \times \mathbf{r} \Rightarrow T = \frac{1}{2}\boldsymbol{\omega}^T I \boldsymbol{\omega} \Rightarrow \mathbf{L} = I\boldsymbol{\omega} \Rightarrow \frac{d\mathbf{L}}{dt} = \mathbf{N} \Rightarrow \text{Euler's equations.}}$$

The key physical insight is:

$$\boxed{\mathbf{L} \text{ is not generally parallel to } \boldsymbol{\omega}.}$$

This is why rigid body rotation displays precession, wobbling, and intermediate-axis instability.